`backend/core/context_manager.py` Documentation (Roman Urdu)

Yeh file LLM chat ke **Context aur Memory** ko manage karne ke liye banai gayi hai. LLM models ki ek maximum limit hoti hai (jaise 4096 tokens). Agar chat history bohot lambi ho jaye toh model bhoolne lagta hai ya crash kar jata hai. Yeh file is maslay ka hal nikalti hai.

File Ka Maqsad (Purpose)

Is file mein 3 main cheezein implement ki gayi hain: 1. **Context Isolation**: Har chat thread ki memory alag rakhna taake messages mix na hon. 2. **MemGPT-style Virtual Paging**: Jaise computer aapki RAM bharne par hard-disk (virtual memory) istamal karta hai, wese hi yeh puranay messages ko hard-disk pe park kar deta hai aur zrurat padne pe waapis (page-in) le aata hai. 3. **Context Compression**: Puranay messages ka khulasa (summarization) karna taake tokens bachein.

Code Blocks aur Classes ki Tafseel

1. `class ContextCompressor`

- **Kya karta hai?**: Yeh class lambay paragraphs ya message history ko compress (chota) karti hai.

  Key Functions:

- `compress_messages(...)`: System aur naye messages ko untouched chorta hai jabke 4 se puranay messages ko compress kar deta hai.
- `_extract_key_sentences(text, ratio)`: Yeh code text ko sentences mein todta hai aur sirf un paragraphs ko rakhta hai jisme zruri lafz hon (like "important", "result", "because"). Baqi faltu batain nikal deta hai.

2. `class MemoryLayerManager`

- **Kya karta hai?**: OS virtual memory system ki tarha active pages (RAM m) aur inactive (Database m) ko sambhalta hai.

  Key Functions:

- `create_pages_from_messages()`: Chats ko chunks (pages) mein todta hai aur limit paar honay pe purane pages discard kar deta hai (is_active=False).
- `page_in_relevant(pages, query)`: Agar user kisi aisi purani baat ka zikar karay jo memory se out ho chuki thi, toh yeh search maar k sirf us related page ko dubara context window (active=True) kar deta hai.

3. `class ThreadContextBuilder`

- **Kya karta hai?**: Yeh aakhri builder file hai jo FastAPI route se request lekar us mein lagne wale messages ko perfectly line up karta hai.

Key Functions:

- `build_isolated_context()`: Har thread ki apni settings dekhta hai ke usme compression enabled hai ya MemGPT mode active hai, aur phir context filter ho kar directly LLM ko bheje jane k kabil ho jata hai.